Predicting EU27 Greenhouse-Gas Emissions from Sector-Level Data "Given the sector-level GHG emissions of EU27 countries in year t, predict the total EU27 emissions in year t+1"

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Abstract

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1 Introduction and Methods

1.1 Context and research question

1.2 Datasets

1.2.1 Toy datasets

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1.3 Methods

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3 Real Dataset Results

3.1 Data preparation

The raw EDGAR booklet contains 4,849 rows, each row describing the annual emissions (Mt CO_2-eq) of a (country, sector, substance) triplet over the period 1970–2024. Before any modelling, the data are aggregated to the EU27 level and transformed so that the eight sectoral features are numerically comparable.

3.1.1 Aggregation to EU27 sector level

The EDGAR dataset already provides a pre-computed `EU27` entity (the 27 current EU member states aggregated according to the Union's official composition), which we adopt to avoid re-implementing the aggregation ourselves. For each of the four substances, we sum the contributions of the eight emission sectors — Agriculture, Buildings, Fuel Exploitation, Industrial Combustion, Power Industry, Processes, Transport, Waste — yielding a clean matrix

$$E \in \mathbb{R}^{T \times S}, \qquad T = 55,\ S = 8,$$

where $E_{t,s}$ is the total CO_2-equivalent emission of sector s in year t. After aggregation, E contains no missing values, which is a key argument for working at the EU27 level rather than at the country level (where several time series are incomplete, especially for the pre-1990 period). The target variable is the annual total

$$y_t \;=\; \sum_{s=1}^{S} E_{t,s}, \tag{1}$$

which drops from $4572\,\mathrm{Mt\,CO_2{-}eq}$ in 1970 to $3165\,\mathrm{Mt\,CO_2{-}eq}$ in 2024 (-30.8%, see Figure 1).

3.1.2 Log-transformation

Raw sectoral emissions span almost an order of magnitude — the Power Industry alone emits over $1300\,\mathrm{Mt\,CO_2{-}eq/yr}$ in the late 1980s, whereas Waste stays below $200\,\mathrm{Mt\,CO_2{-}eq/yr}$ for the entire period. Such heterogeneity makes a regularised regression unfair: the penalty term

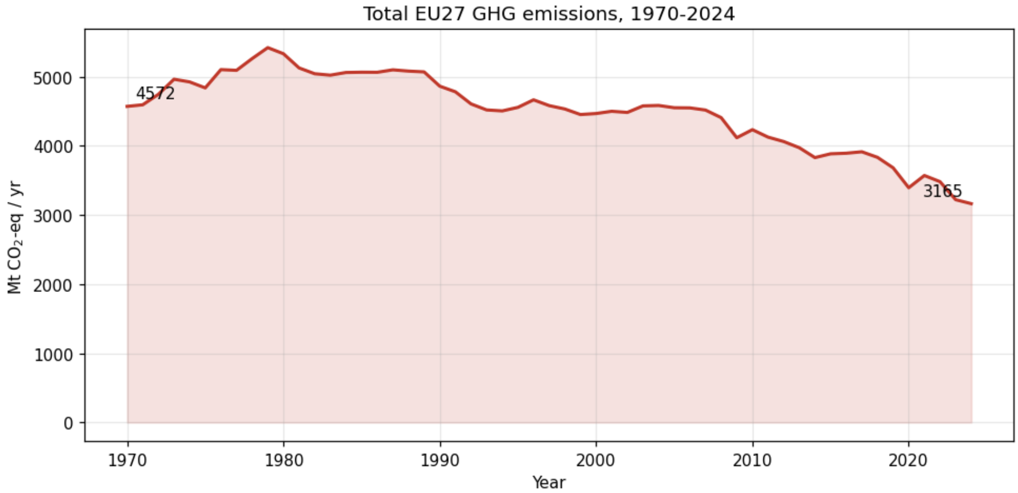

Figure 1: Total EU27 greenhouse-gas emissions, 1970–2024. Aggregated from EDGAR 2025 across the eight sectors and four substances.

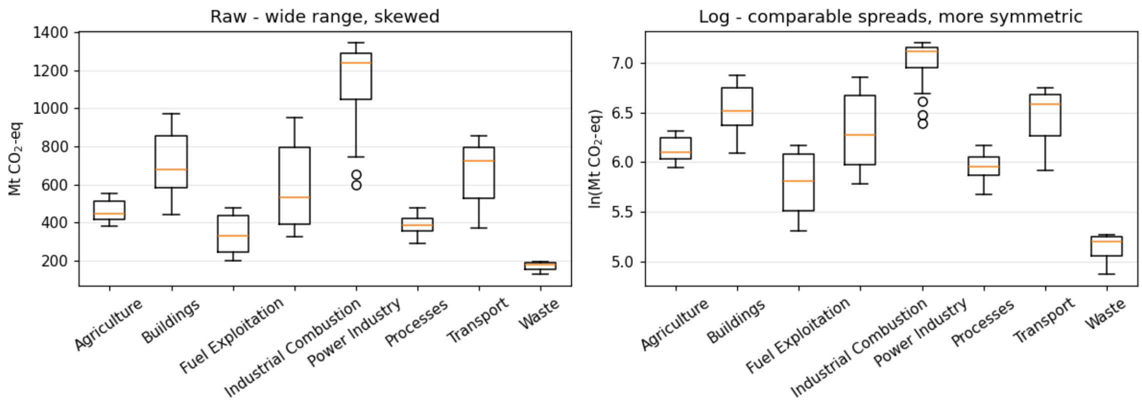

Figure 2: Distribution of sectoral emissions before (left) and after (right) the log-transformation. The log fixes the right-skew and brings sectors onto comparable spreads.

$\lambda\|\beta\|$ would mechanically shrink the small-magnitude features more aggressively. Because all entries of E are strictly positive, we apply a natural logarithm element-wise,

$$\tilde{E}_{t,s} \;=\; \ln E_{t,s}, \qquad \tilde{y}_t \;=\; \ln y_t, \tag{2}$$

which compresses the dynamic range from a factor of ~ 7 to a factor of ~ 1.5 and stabilises the variance (Figure 2). It also linearises any underlying exponential growth or decay, which makes the resulting linear regression interpretable as a model of relative (percentage) changes.

3.1.3 Standardization

After the log-transform, the eight features still have different means ($\overline{\tilde{E}_s}$ ranges from ~ 5.1 for Waste to ~ 7.0 for Power Industry). To put them on the same scale, we apply a z-score normalisation

$$z_{t,s} \;=\; \frac{\tilde{E}_{t,s} - \mu_s}{\sigma_s}, \qquad \mu_s = \frac{1}{T}\sum_t \tilde{E}_{t,s}, \quad \sigma_s^2 = \frac{1}{T}\sum_t \left(\tilde{E}_{t,s} - \mu_s\right)^2. \tag{3}$$

After standardisation each feature has zero mean and unit variance, so Ridge and Lasso penalties act symmetrically on every sector (Figure 3, right panel).

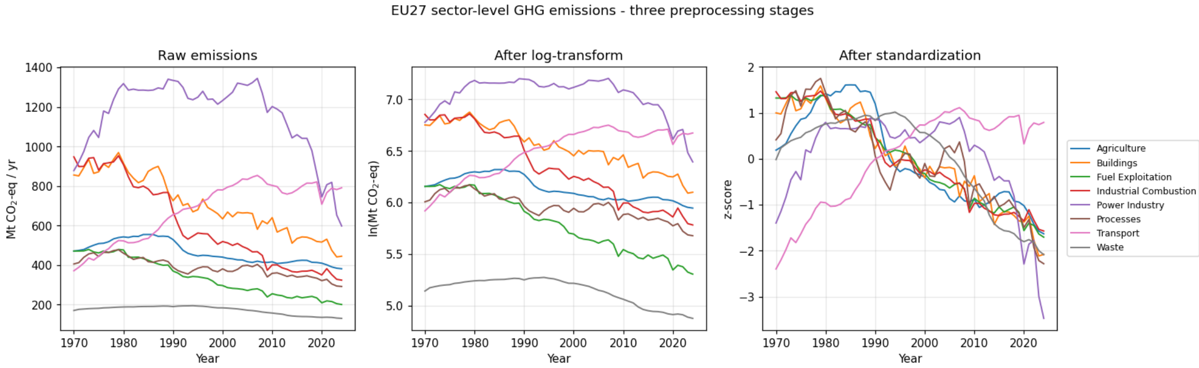

Figure 3: Sectoral EU27 emissions at the three stages of preprocessing: raw (left), after natural logarithm (centre), and after z-score standardisation (right). The transformations preserve the temporal patterns while equalising the numerical scale of the eight features.

A note on data leakage. For the final modelling, μ_s and σ_s are recomputed from the *training fold only* and then applied unchanged to the validation/test folds; otherwise the test statistics would influence the scaling. The global statistics shown in Figure 3 are used only for exploratory visualisation.

3.1.4 Supervised learning setup

To address the research question *"given the sector-level emissions in year t, predict the total emissions in year $t+1$"*, we build a one-step-ahead supervised dataset. Each predictor row is a vector of the eight log-transformed sector emissions, and the corresponding target is the next-year log-total:

$$\mathbf{x}_t = (\tilde{E}_{t,1}, \ldots, \tilde{E}_{t,S}) \in \mathbb{R}^8, \qquad y'_t = \tilde{y}_{t+1} \in \mathbb{R}, \qquad t = 1970, \ldots, 2023. \tag{4}$$

This yields $N = 54$ paired observations $\{(\mathbf{x}_t, y'_t)\}_{t=1970}^{2023}$, with predictors covering 1970–2023 and targets covering 1971–2024. Table 1 summarises the pipeline.

Table 1: Summary of the preprocessing pipeline for the real-world dataset.

Stage	**Operation**	**Resulting shape**
Raw EDGAR booklet	Filter `EU27`, 4×8 − 7 rows	25×55
Aggregation	Sum substances per sector (Equation (1))	$E \in \mathbb{R}^{55\times 8}$
Log-transform	Natural log of E and total (Equation (2))	$\tilde{E}$, $\tilde{y}$
Standardization	z-score on training fold (Equation (3))	z, $\tilde{y}_{\text{std}}$
Supervised pairs	$(\mathbf{x}_t, y'_t) = (\tilde{E}_t, \tilde{y}_{t+1})$ (Equation (4))	$N = 54$

3.2 Clustering of countries

The main objective of this project is to predict the total greenhouse gas emissions of the EU27. However, the aggregated EU dataset only provides 54 years of historical data ($N = 54$). Training a regression model on such a small dataset can be difficult and may lead to overfitting or poor predictions. To solve this problem, an unsupervised learning step was introduced before the main regression task. The goal of this clustering step is to analyze the 194 global countries and group them by their industrial and economic profiles. By identifying which countries around the world share the same emission patterns as the European member states, the training dataset can potentially be expanded with similar global countries to build a stronger and more accurate prediction model.

3.2.1 Data Preparation for Clustering

Before grouping the 194 countries from the EDGAR dataset, the data had to be prepared differently than for the regression. The goal was to group countries by what their economy looks like today, not by how big they are or their past emissions. So, only recent years were used.

First, big groups like "EU27", "GLOBAL TOTAL", and international transport zones were removed. Then, all the emissions for each sector were added up and the average for the last 15 years (2010–2024) was calculated.

To make sure the computer looks at the shape of the emissions and not the total size of the country, the numbers were changed into percentages (like Sector Emissions / Total Country Emissions). Finally, a z-score standardization (StandardScaler) was applied. This was done because some sectors (like Waste) are small and do not change much. It was important to make sure these small sectors were just as important for the K-Means as big sectors like the Power Industry.

3.2.2 K-Means Optimization and Silhouette Analysis

After the data was ready, K-Means was used to find groups of countries with similar emissions. To find the best number of groups (K), the Silhouette Score was checked for K from 2 to 10.

The best score was 0.2453 when $K = 8$. A score of about 0.25 means some groups overlap a little bit, but it still shows a real difference between the country profiles. So, $K = 8$ was chosen for the final model.

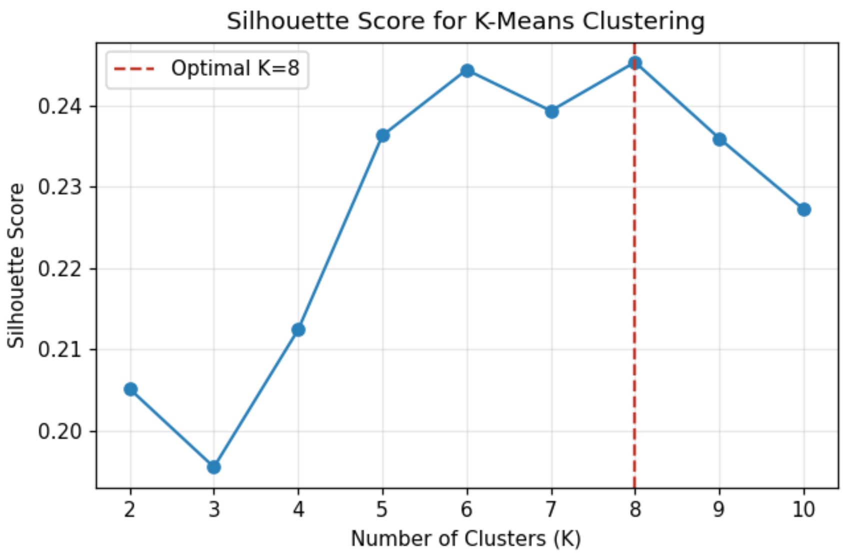

Figure 4: Silhouette scores for different numbers of clusters (K). The peak at $K = 8$ indicates the optimal mathematical separation of global emission profiles.

3.2.3 The Fragmentation of the EU27

A very important result from this step is that the EU27 countries are not all in the same group. At first, it seemed like the EU would have one single profile. But the countries have very different economies, so the EU27 is actually split into three different groups.

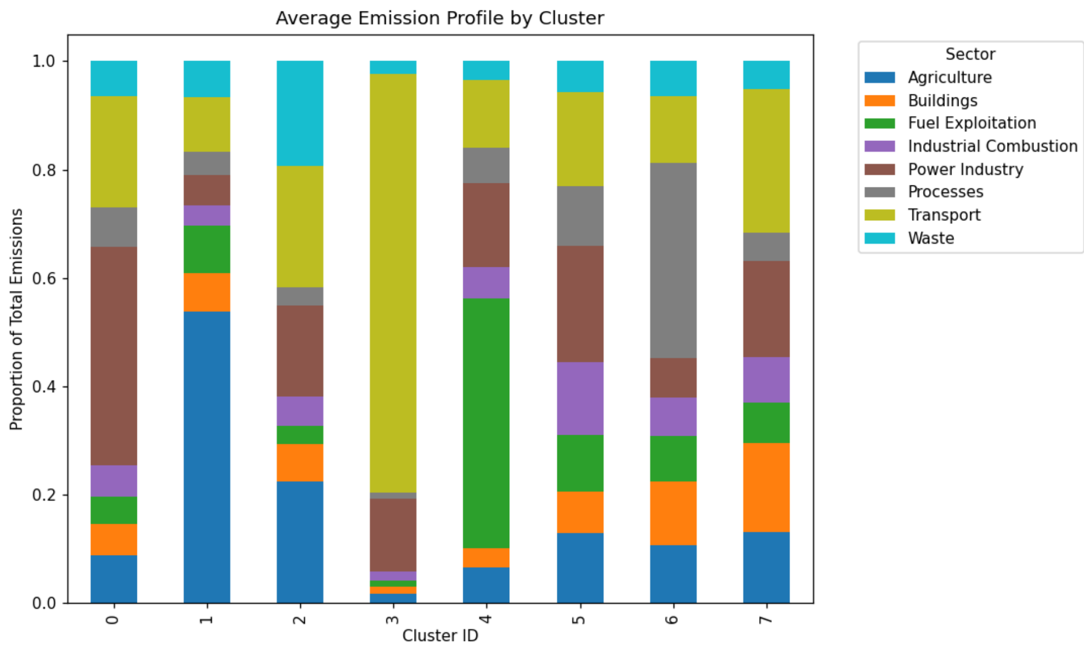

Figure 5: Average sectoral emission profiles for the three clusters containing EU27 member states. Values represent the percentage contribution of each sector to the country's total emissions.

The 27 countries are in Cluster 0 (7 countries), Cluster 5 (10 countries), and Cluster 7 (10 countries). Here is what these groups look like:

- **Cluster 0 (High Power Industry):**
  - *Members (7):* Bulgaria, Cyprus, Czechia, Estonia, Greece, Malta, Poland.
  - *Profile:* These countries have a lot of emissions from the Power Industry (about 40.3% on average). This means they probably still use a lot of coal and fossil fuels for their electricity.
- **Cluster 5 (Balanced Industry):**
  - *Members (10):* Austria, Croatia, Finland, Lithuania, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden.
  - *Profile:* This group is very balanced. They have Power Industry emissions (21.5%), but also higher emissions from Industrial Combustion (13.4%) and Agriculture (13.0%) than the other European countries.
- **Cluster 7 (Transport and Buildings):**
  - *Members (10):* Belgium, Denmark, France, Germany, Hungary, Ireland, Italy, Latvia, Luxembourg, Netherlands.
  - *Profile:* Most of their emissions come from Transport (26.6%) and Buildings (16.5%). Their Power Industry is lower (17.9%). This is normal for rich countries that have cleaner electricity but lots of cars and need a lot of heating for houses.

3.2.4 Implications for Supervised Modelling

Because there is no single "EU27 group," there is a problem for the next step (the regression). The total EU emissions are actually a mix of three very different types of economies. So, the model cannot just be trained on one "global EU" group anymore.

For the supervised step, the plan has to change. Either (a) three different models must be made for the three groups and the results added together, or (b) a new training dataset must be created that mixes the three groups in the same 7:10:10 ratio to predict the total EU27 emissions.

3.3 Predicting EU27 emissions

3.4 Forecast 2025–2034

4 Discussion and Conclusions

AI Disclosure

We use Claude, ChatGPT, GitHub Copilot, etc for these tasks ...

References

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A Appendix